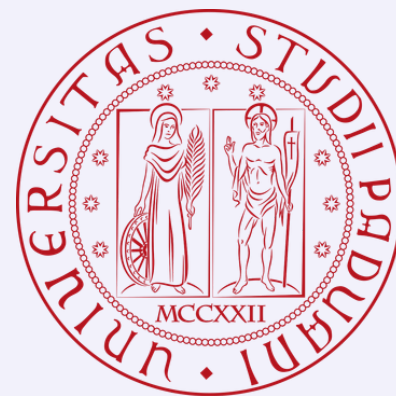

Task 1: Source Retrieval for Scientific Web Claims

Search Engines 2025/2026 – DEI, University of Padua – Master degree in Computer Engineering
SEUPD@CLEF: Team RETRIX



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The project



CHECKTHAT! LAB AT CLEF 2026

[Home](#)[Editions](#)[Tasks](#)

Contents

- [Definition](#)
- [Datasets](#)
- [Evaluation](#)
- [Submission](#)
- [Leaderboard](#)
- [Organizers](#)

Task 1: Source Retrieval for Scientific Web Claims

Definition

Source Retrieval for Scientific Web Claims: Given a social media post that contains a scientific claim and an implicit reference to a scientific paper (mentions it without a URL), retrieve the mentioned paper from a pool of candidate papers.

The data comprises English, German, and French social media posts.

Goal: Source Retrieval for Scientific Web Claims

Scale: 10000 papers, ~19000 topics to train on

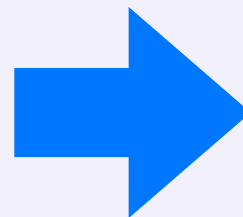
Input: Short, noisy social media posts in English, French, and German

Target metric: MRR@5

The challenges

- **The linguistic gap:** academic papers vs colloquial tweet-style queries
- **Implicit references:** semantic clues instead of proper citations.
- **Multilinguality:** queries in EN/FR/DE, papers in EN only

Presymptomatic SARS-CoV-2 Cases
and Spread in a Nursing Home |
NEJM #ACF #COVID19



Presymptomatic SARS-CoV-2 Infections and Transmission in a Skilled Nursing Facility

Background Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection can spread rapidly within skilled nursing facilities. After identification of a case of Covid-19 in a...

Our baseline

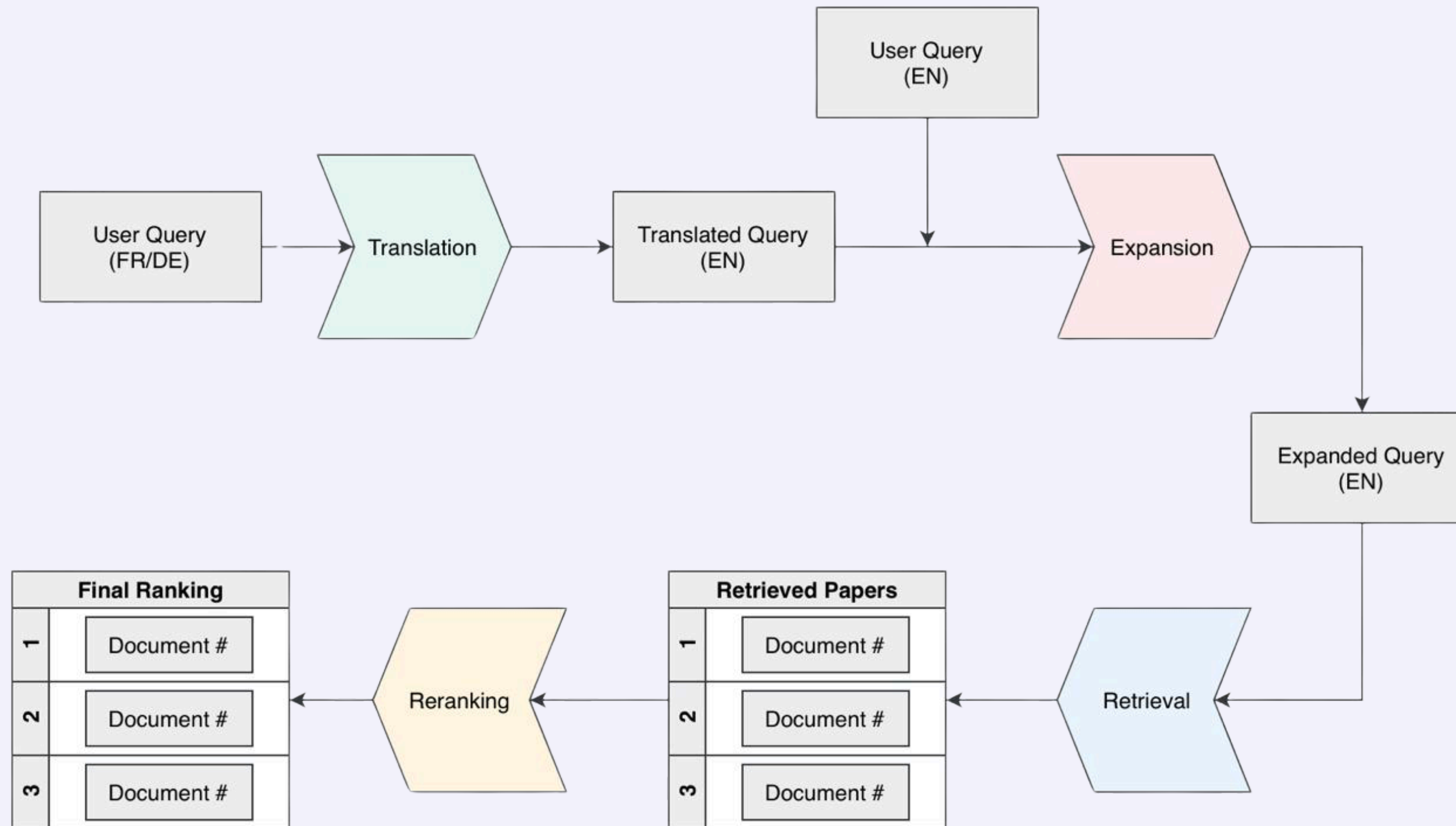
We used Lucene BM25 as the baseline

Best configuration:

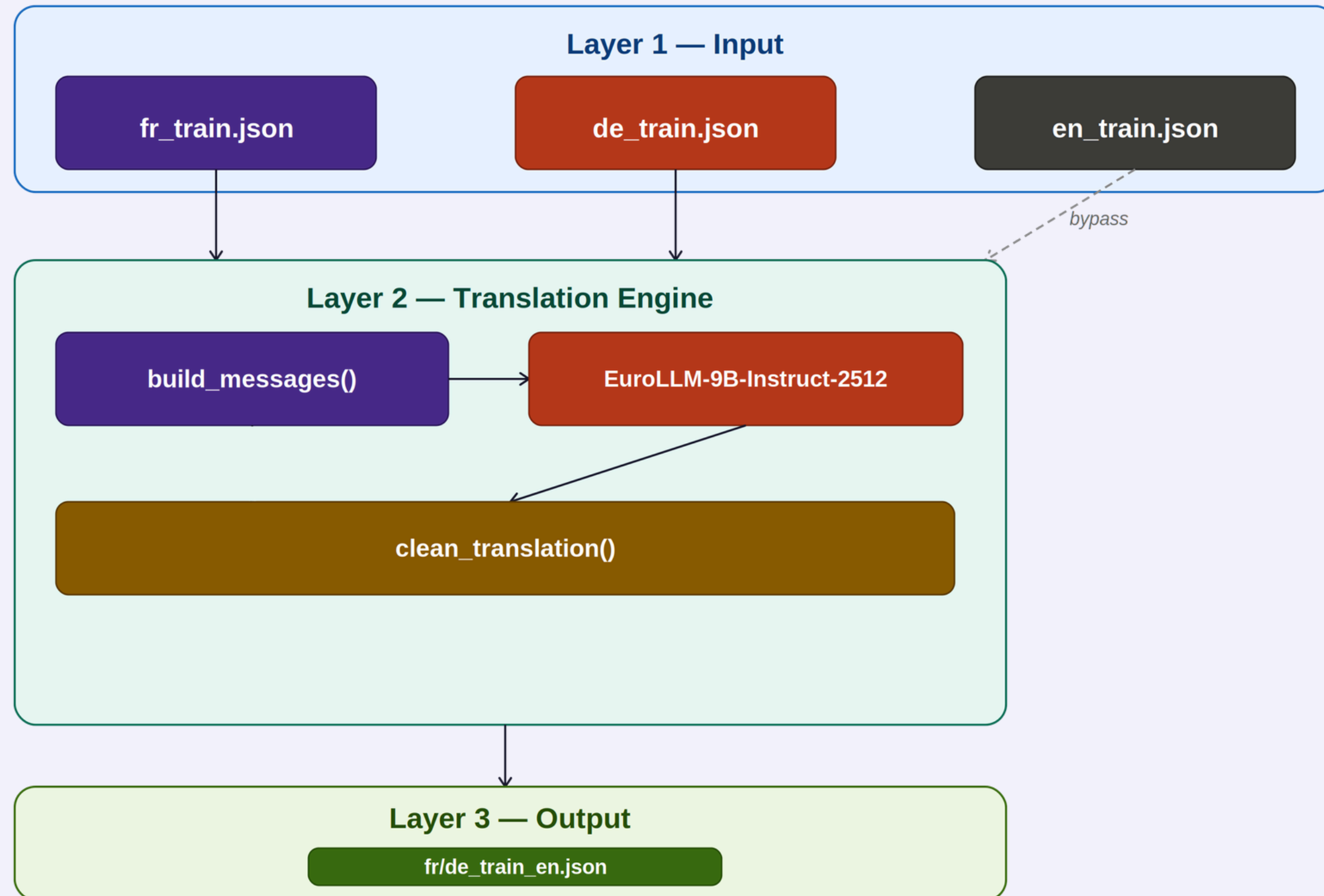
- domain-targeted stoplist of ~670 words
- trim filter, lowercase filter
- ASCIIFoldingFilter, EnglishPossessiveFilter, kStemFilter
- no WordDelimitedGraphFilter
- removal of “#” but not “@”
- BAAI/bge-large-en-v1.5 query expansion
- MRR@5 = 0.503 and Recall@100 = 0.796 on en_train dataset

Final pipeline: MRR@5 = 0.702 and Recall@100 = 0.911

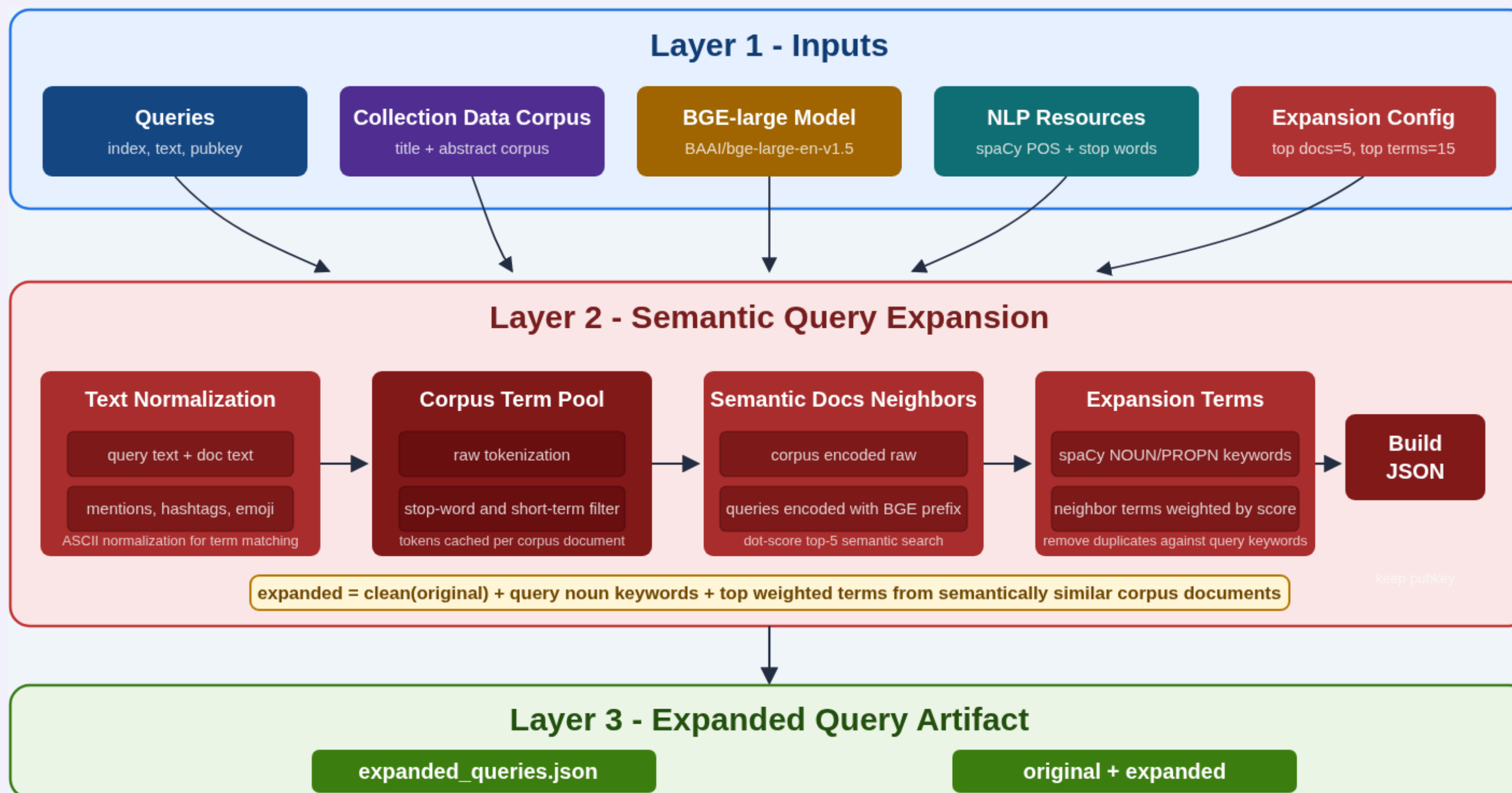
Overview of our pipeline



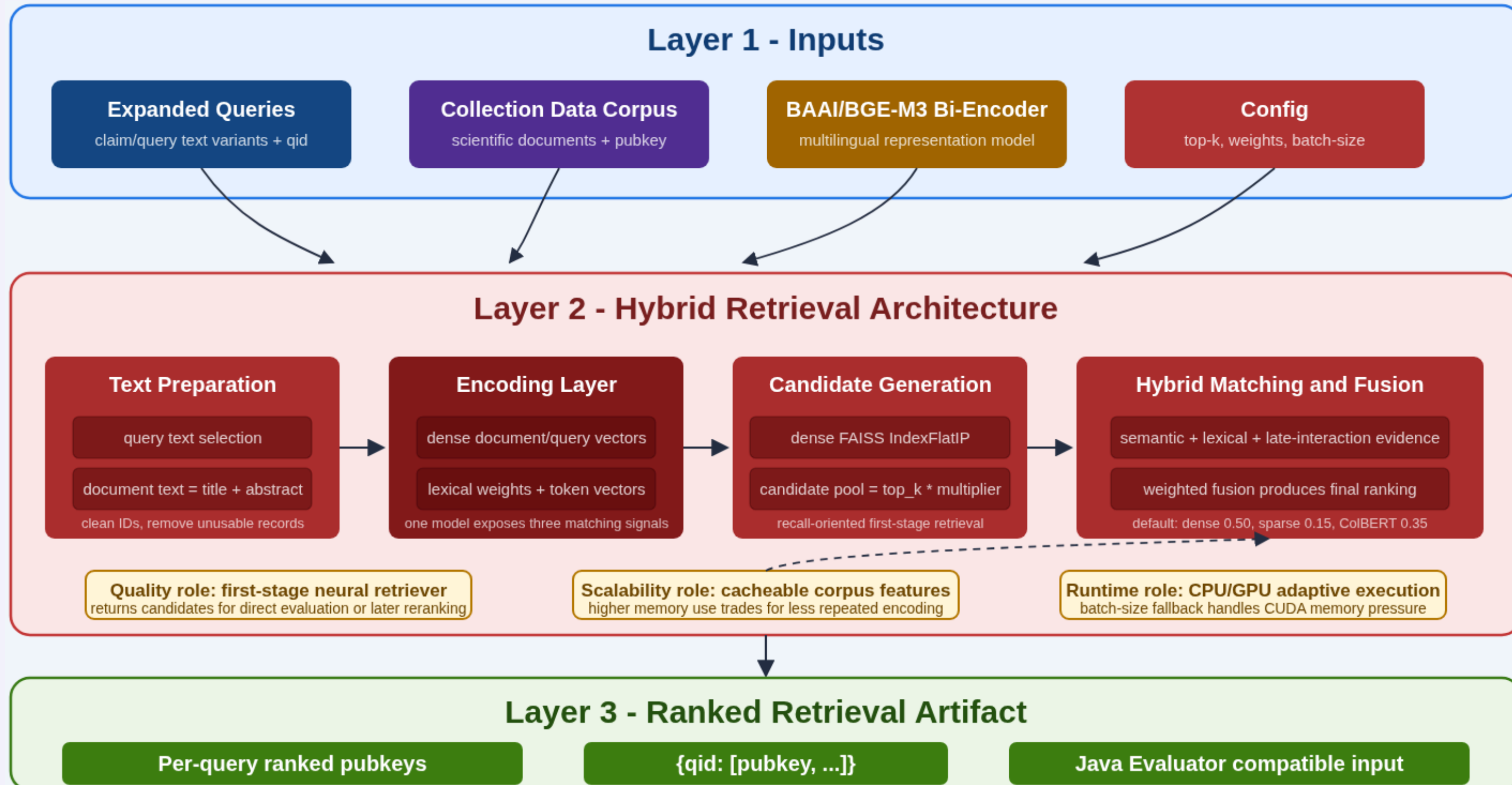
Query Translation



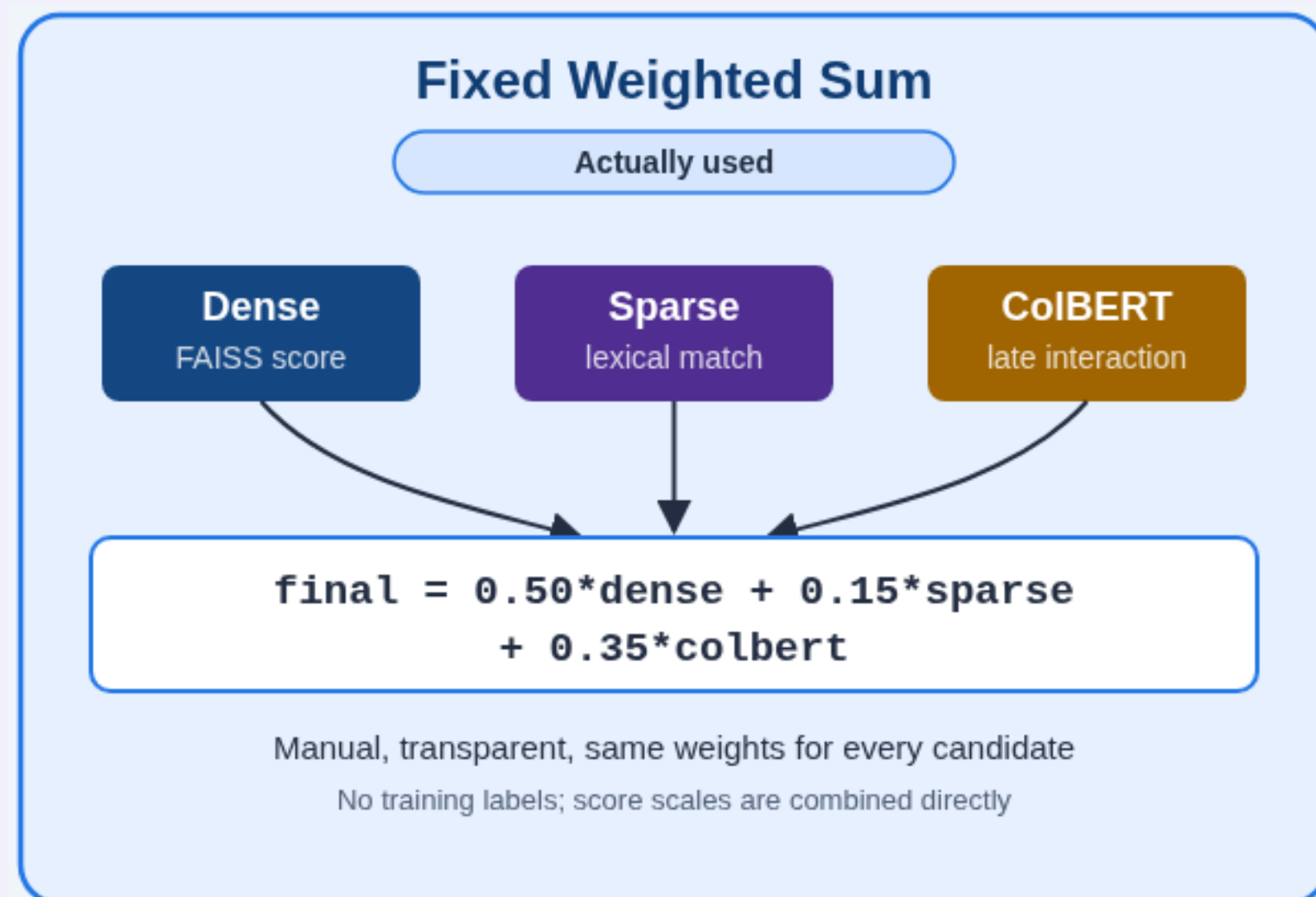
Query Expansion



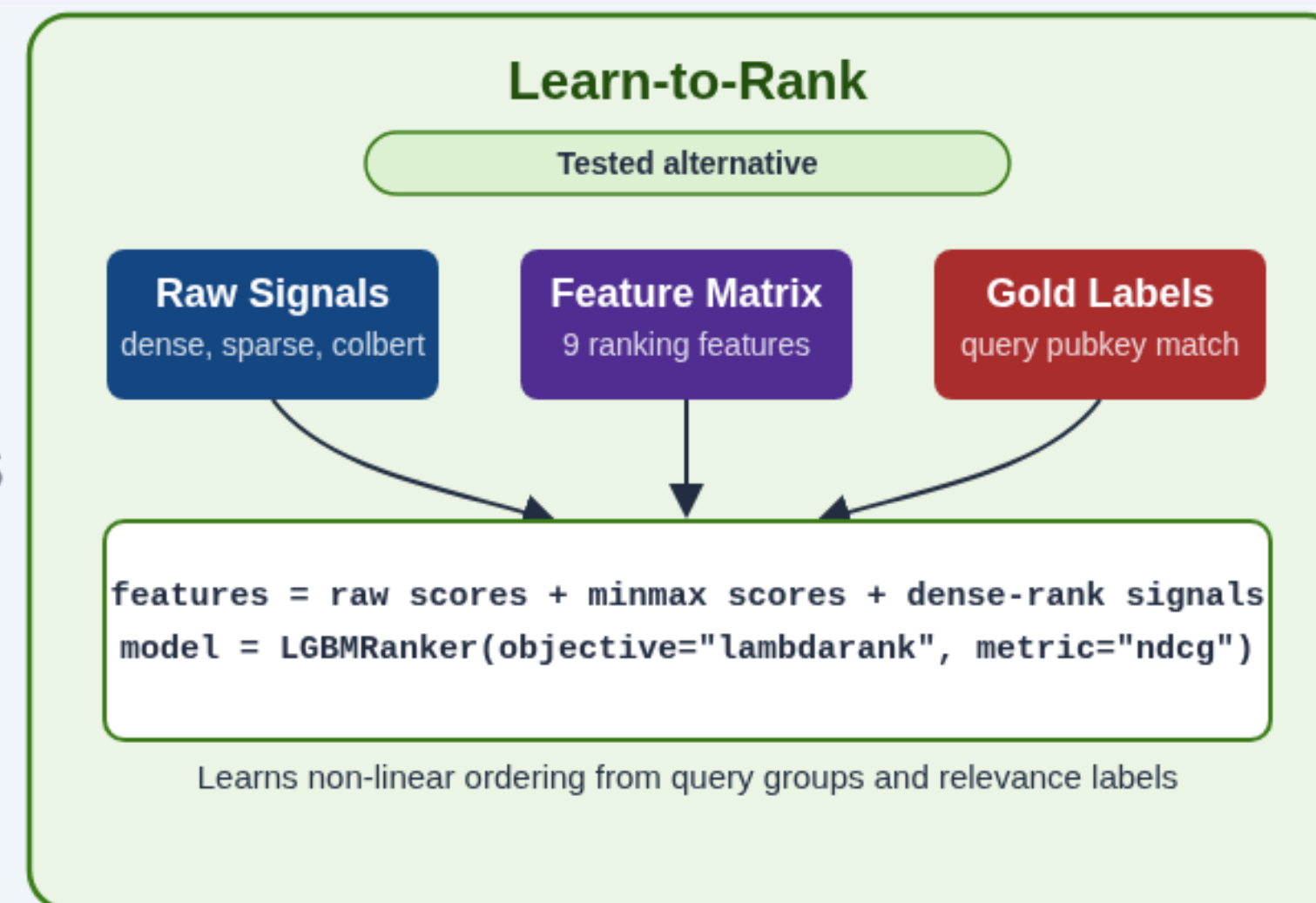
Retrieval



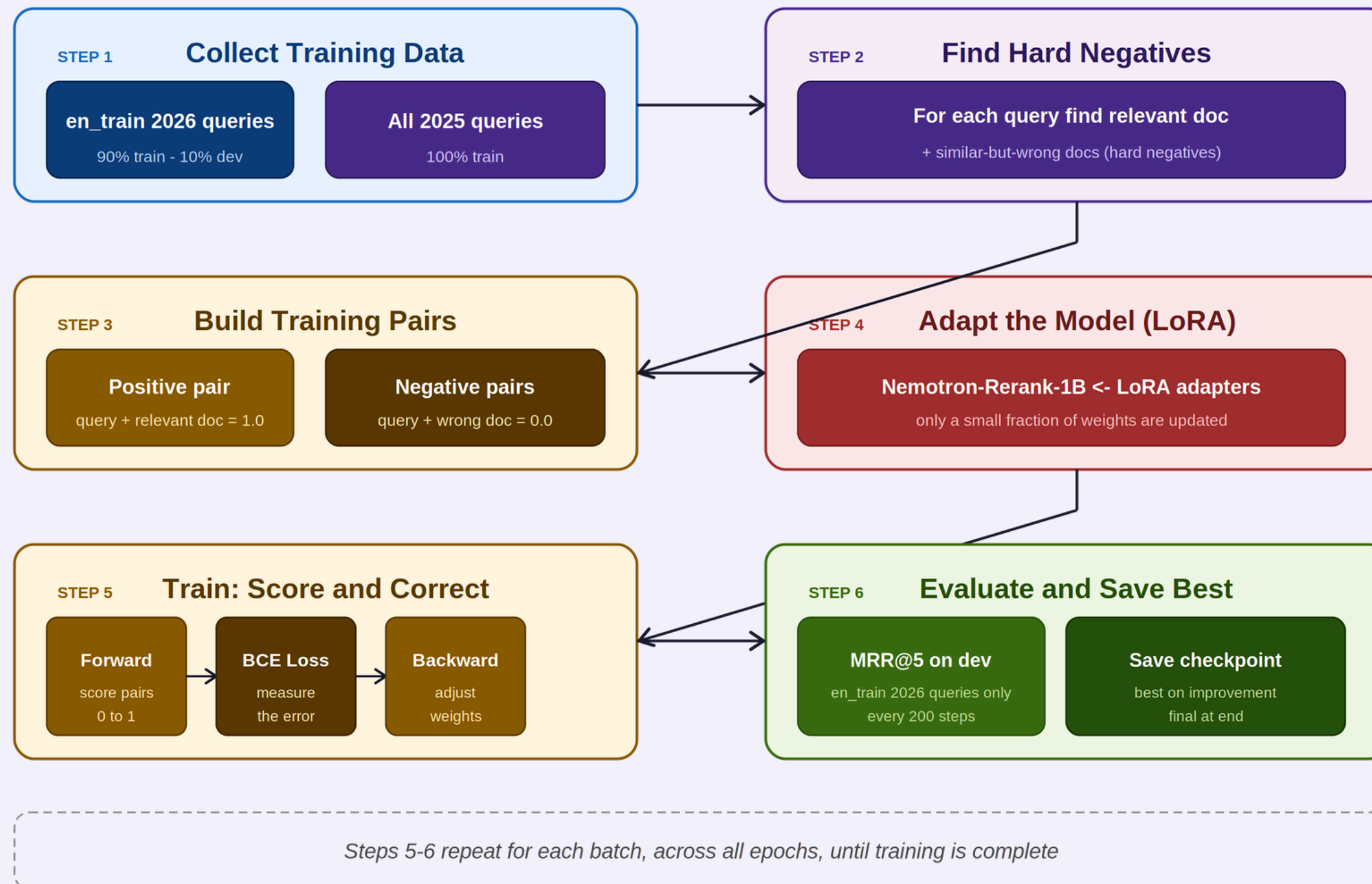
Scoring



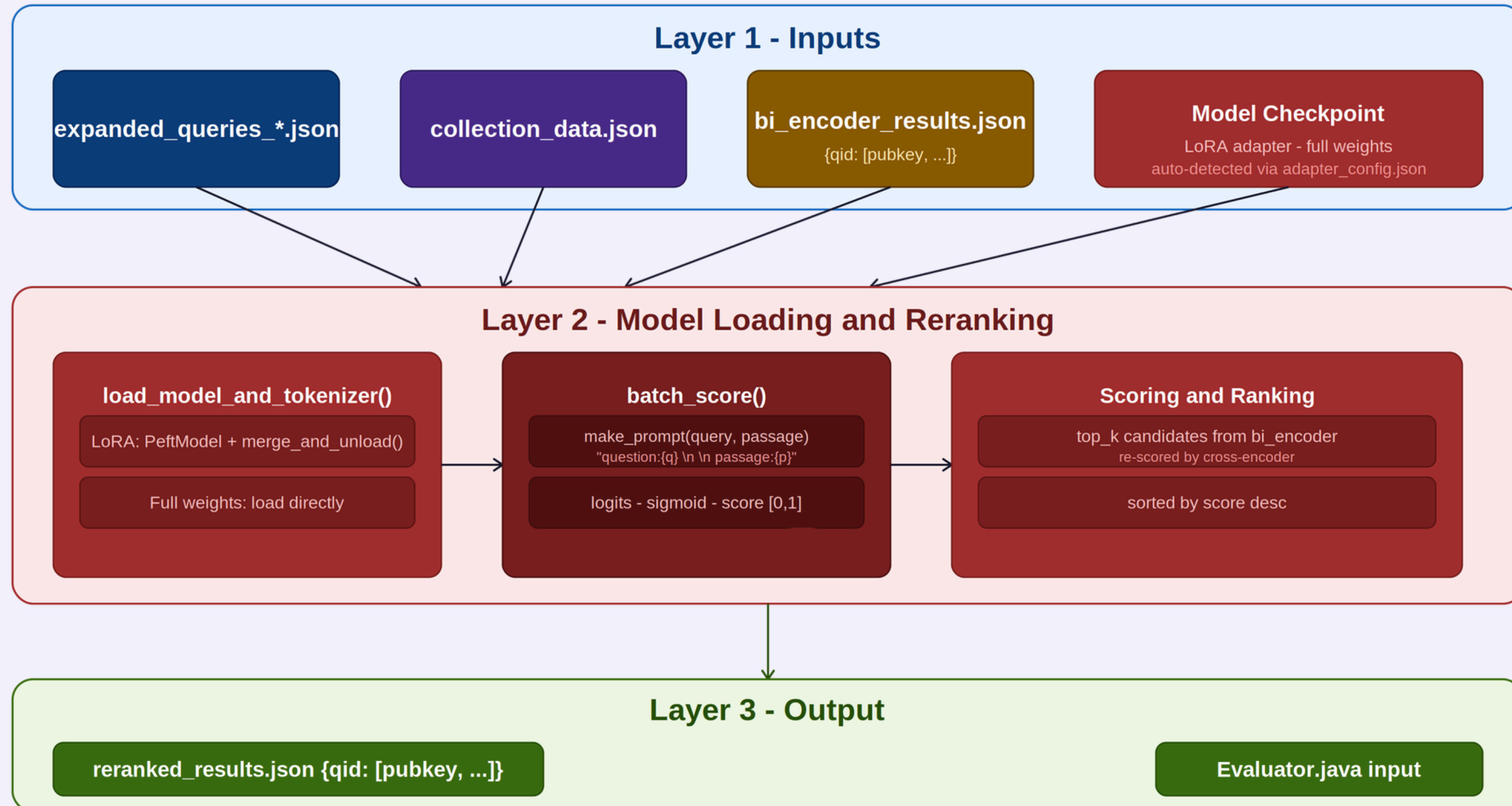
VS



Fine-tuning



Reranker



Statistical Analysis

Per-query scores

Stage-aligned metrics:

- **MRR@5** (early precision)
- **Recall@100** (candidate coverage)

Friedman omnibus test

Pairwise Wilcoxon signed-rank tests with Holm correction

Friedman Omnibus Tests

Goal: Detect overall differences across systems

H_0 : all tested systems are equivalent in terms of per-query MRR@5.

- English TRAIN split
- 14,977 labeled queries
- $\chi^2 = 21368.46$
- $p < 0.001$
- Reject H_0

Friedman does not identify which systems differ: pairwise tests are needed next.

Holm-Corrected Pairwise Wilcoxon Tests

Systems / Comparison	Mean difference	p_{holm}	Interpretation
H315 vs <i>H513</i>	– 0.0005	1.0000	No significant MRR@5 difference
H424 vs H513	+ 0.0500	8.58×10^{-116}	H513 preferred for candidate coverage
H513 vs N1000	+ 0.1451	< 0.001	N1000 significantly better
N1000 vs N500	– 0.0008	0.0497	Statistically significant, practically negligible

Rank-Bucket & Error Analysis

Split	Family	Queries	Rank 1	Ranks 2-5	Ranks 6-10	Ranks 11-100	Rank >100	Missing
EN TRAIN	H513 (topk100)	14977	7459 (49.8%)	2280 (15.2%)	715 (4.8%)	2114 (14.1%)	0 (0.0%)	2409 (16.1%)
EN TRAIN	N1000	14977	9697 (64.7%)	2057 (13.7%)	598 (4.0%)	1291 (8.6%)	405 (2.7%)	929 (6.2 %)
EN DEV	N1000 Fine-Tuned	3905	2574 (65.9%)	538 (13.8%)	144 (3.7%)	309 (7.9%)	99 (2.5%)	241 (6.2%)
EN TEST	N100 Fine-Tuned	6076	3863 (63.6%)	821 (13.5%)	157 (2.6%)	245 (4.0%)	0 (0.0%)	990 (16.3%)
EN TEST	N2000 Fine-Tuned	6076	3950 (65%)	926 (15.2%)	229 (3.8%)	506 (8.3%)	0 (0.0%)	465 (7.7%)

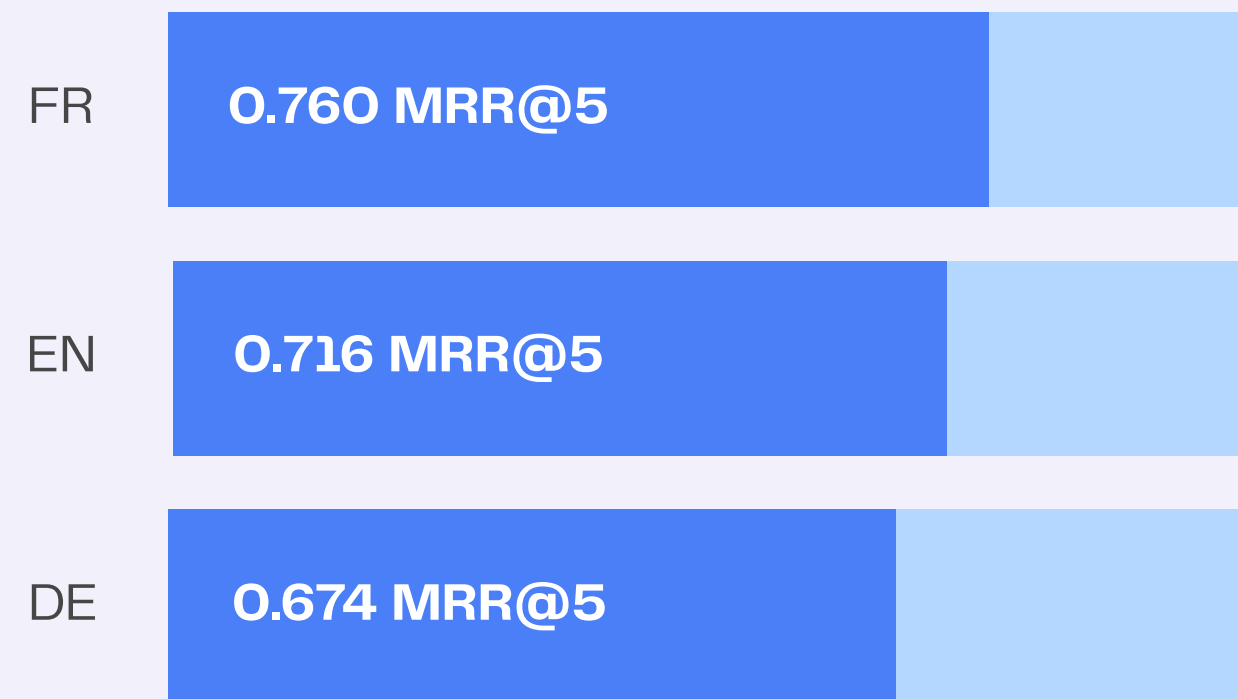
Runs comparison: TRAIN Split

Key Progression

- Lexical Gap: BM25 establishes a solid baseline but suffers from vocabulary mismatches;
- Hybrid First-Stage: BGE-M3 combines dense & ColBERT vectors to secure candidate recall;
- Reranking Edge: Nemotron fine-tuned via LoRA shifts the median MRR@5 from 0.5 to 1.0.

Configuration	MRR@5	Recall@100	nDCG@10
Lucene BM25 (AnalyzerV2)	0.553	0.796	-0.054
BGE-M3 Hybrid (H513)	0.557	0.839	Baseline
Nemotron top 100	0.685	0.839	+0.128
Nemotron top 1000	0.702	0.911	+0.145

Multilingual Validation on DEV Split



Multilingual insight :

- Cross-Lingual Success: Translation-based query processing proves robust, with French yielding the strongest scores;
- German Terminology Drift: German exhibits wider variance and a drop in early precision;
- Consistent Recall: All three languages maintain high Recall@100, validating the BGE-M3 candidate generation pool.

Performance by Query Features

QUERY LENGTH vs MRR@5

SHORT (<21 token)

0.760 MRR@5



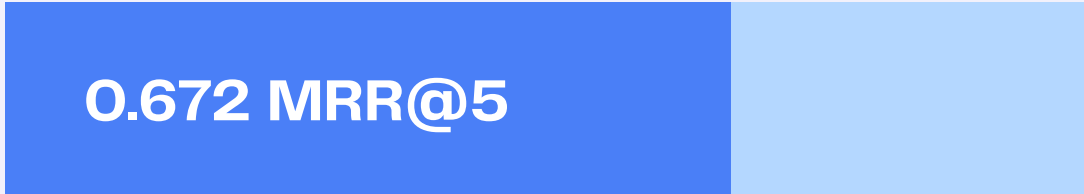
MEDIUM (21–34 token)

0.713 MRR@5



LONG (>34 token)

0.672 MRR@5



NAMED-ENTITY RICHNESS vs MRR@5

ENTITY RICH (Authors, Orgs)

0.733 MRR@5



ENTITY SPARSE

0.664 MRR@5



Failure Mode

METADATA MISMATCH

some claims, like those mentioning specific numbers or numerical results which are not mentioned in the abstract of the papers, are found hard to retrieve.

TRANSLATION DRIFT

literal translation, especially from German may introduce some terms reducing lexical overlap with the English corpus (e.g. for the numerous compound words German is famous for).

IMPLICIT AND CONVERSATIONAL QUERIES

Vague references lack concrete keywords, highlighting the crucial need for dense semantic dual-encoders over lexical search alone.

Conclusions

Combining EuroLLM translation, embedding-based expansion, and BGE-M3 hybrid retrieval establishes a highly competitive candidate pool.

Fine-tuning Nemotron-1B using LoRA parameter-efficient adapters provides a definitive performance leap, moving the median MRR@5 to 1.0.

FUTURE WORK

- Explore Data Augmentation techniques
- Incorporate full-text bodies or extended metadata
- Deploy domain-adapted translation models

Thank you